

END TERM EXAMINATION
SECOND SEMESTER [B.TECH] MAY-JUNE 2026

Paper Code: ES-114

Subject: Engineering Mechanics

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

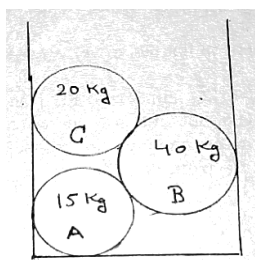
Q1. Attempt **any four** of the following questions:

(4×5=20)

- (a) State and prove Varignon's theorem.
- (b) Define dry friction. State laws of dry friction.
- (c) Prove that two elastic bodies of equal masses exchange velocities in case of direct central impact.
- (d) Explain various types of beams.
- (e) Explain conservation of energy and conservation of momentum.

Q2. Three cylinders A, B, and C of radius 4 cm, 6 cm and 5 cm respectively are piled up in a rectangular channel as shown in Diagram-I. Determine the reaction between cylinder A and the vertical wall of the channel.

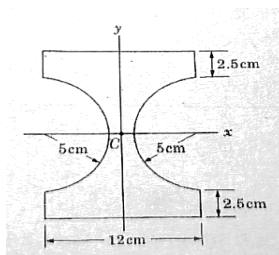
(10)



OR

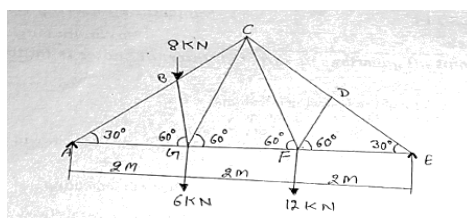
Q3. Find the moment of inertia of the cross section of iron beam with respect to the centroidal axis (Diagram-II).

(10)



Q4. Determine the nature and magnitude of the forces in the members BC, GC and GF of the truss shown in Diagram-III.

(10)



OR

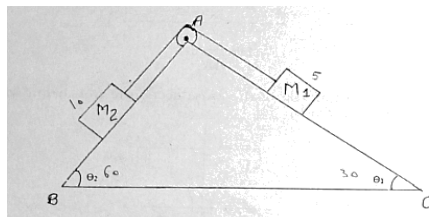
Q5. Derive the ratio of tensions in a flat belt drive.

(10)

P.T.O.

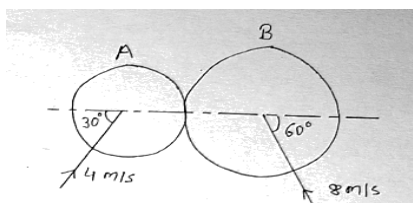
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- Q6.** Two blocks of mass $M_1 = 5 \text{ kg}$ and $M_2 = 10 \text{ kg}$ placed on two inclined planes of elevation $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$ are connected by a string as shown in Diagram-IV. Find the acceleration of the masses. The coefficient of friction between the blocks and the plane is 0.33. (10)

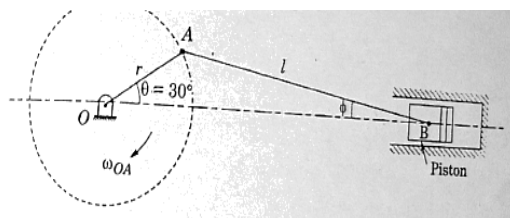


OR

- Q7.** Two smooth spheres A & B having a mass of 2 kg and 4 kg respectively collide with initial velocities as shown in Diagram-V. If the coefficient of restitution for the spheres is $e = 0.8$, determine the velocities of each sphere after the impact. (10)



- Q8.** A reciprocating engine mechanism is shown in Diagram-VI. The crank OA has a constant angular velocity of $N = 300 \text{ rpm}$. The crank OA is of length $r = 12 \text{ cm}$ and connecting rod AB of length $l = 60 \text{ cm}$. Find (a) the angular velocity of the connecting rod (b) the velocity of the piston when the crank makes an angle $\theta = 30^\circ$ with the horizontal. (10)



OR

- Q9.** Draw the shear force and bending moment diagram for a cantilever beam loaded as shown in Diagram-VII. (10)

